

SOLUTION

NaviFreight is an end-to-end maritime intelligence and multi-modal supply chain decision-support system engineered for Indian public sector steelmakers (SAIL) to dynamically optimize coal chartering, berth scheduling, rail-truck dispatch, and blast-furnace stockyard resilience.

- Optimal Market Entry & Tender Timing:** Advises SAIL on strategic tender booking windows using P10/P50/P90 market forecasts, dynamically recommending the optimal charter ratio between long-term Contracts of Affreightment (COA) and spot fixtures (e.g., maintaining a baseline ~70% COA to ~30% spot mix, shifting with market volatility).
- Vessel Bunching & Furnace-Critical Berthing:** Prioritizes berth access for bulkers feeding coal-starved steelworks (e.g., Rourkela, Bokaro) while eco-pacing non-critical vessels; evaluates whether to divert to alternate ports or hold at outer anchorage based on total demurrage versus fuel and operational costs.
- Railway Rake Shortages & Evacuation Costs:** When FOIS rakes are unavailable at port sidings, plants face expensive road trucking; at 80 NM out, NaviFreight pre-calculates rail vs. truck costs, evaluating bunker fuel to suggest an alternate port if fuel is ample, or the nearest port/trucking if fuel is low.
- Vessel Type Recommendation:** NaviFreight recommends the best-fit vessel for SAIL based on cargo volume and port constraints, using a What-If Simulator to assess alternative ports and identify flexible vessel types based on LOA, beam, and draft compatibility.
- CAG Inventory Management & Carrying Cost:** NaviFreight addresses CAG-identified raw-material management issues by forecasting SAIL's future material requirements and synchronizing vessel arrivals with plant stock levels, helping prevent raw-material shortages, excess inventory, and unproductive vessel time.
- Global Maritime Intelligence & Early Warning Radar:** Tracks Baltic Dry and Capesize indices, S&P Global Platts and Argus coal price assessments, and real-time alerts across critical chokepoints (Red Sea, Malacca Strait) and Australian loading terminals (Gladstone, Hay Point) to predict freight shocks weeks in advance.
- Coastal Triangulation & Domestic Backhaul Pairing:** Analyzes draft clearance to pair discharging import bulkers with domestic coastal cargo movements (such as moving thermal coal from Paradip/Dhamra to southern power plants), eliminating unproductive ballast deadheading.

We solve the problem with an integrated, multi-modal supply chain platform tailored for Indian PSUs, combining predictive logistics, inventory security, and automated coordination. **Operational modules and prototype validation are active.**

WHY WE STAND OUT ?

Our AI solution digitalizes the maritime supply chain, making it dynamic for uninterrupted plant feeding, demurrage prevention, and real-time updates.

**Predictive Simulation & Tender Timing Engine**

Simulates forward freight curves and P10/P50/P90 distributions, advising SAIL on tender release and optimal ~70:30 COA-to-spot ratios.

**Rail Rake & Multi-Modal Siding Evacuation**

Pre-empts port railway shortages by evaluating rail vs. truck freight costs at 80 NM out, eliminating expensive emergency road haulage.

**Blast Furnace Stockyard Protection**

Synchronizes vessel arrivals directly with daily plant consumption, preventing the furnace-deficit shutdowns highlighted in CAG audits.

**Dynamic ETA & Vessel Bunching Prevention**

Synchronizes vessel arrival ETAs via AIS eco-pacing and berth slot allocation, eliminating outer anchorage queues and port demurrage.

ARCHITECTURE & VALUE (NAPKIN FLOW)

